

Sink-float prototype testing - Blantyre, Malawi

Testing Document

Version 1

Date:

- Start: 08.09.2025
- End: 12.10.2025

Contents

Sink-float prototype testing - Blantyre, Malawi	1
General description and purpose.....	3
Purpose of test.....	3
Description of Test.....	3
Calculations.....	4
Schedule of test with all steps required.....	4
Risk analysis	6
Potential risks associated with this test	6
Mitigation steps.....	6
Risks after taking mitigation steps	6
Tools and devices required	6
Testing Results.....	8
Fit Test – 08.09.2025	8
Leakage Test – 08.09.2025	8
Functionality Test – 23.-24.09.2025	8
Plastic flakes acquisition from storage.....	8
Flake washing	8
Plastic separation by sink-float prototype	9
Sample Composition Analysis – 12.10.2025	9
Sample 1.....	10
Sample 2.....	10
Sample 3.....	11

Sample 4.....	11
Sample 5.....	12
Evaluation	12
Bibliography.....	13

General description and purpose

Purpose of test

Prototype tests fulfill two objectives:

1. To verify the satisfactory functionality of individual attributes of the machine. These tests are conducted during the manufacturing process to allow for timely modification in case of failure.
2. To validate the performance of the prototype, its functionality is tested after complete assembly.

In the case of the sink-float separation prototype, key features reviewed during the manufacturing are the tolerances between moving parts (trays and basins) and the leakage integrity of the basins. By ensuring that all trays fit in both basins in any orientation (horizontal rotation), convenient setup and operation is provided. The same applies to leak-proof vessels, which allow the prototype to be set up independently of a drainage in the location of use.

Once construction is complete, the prototype's operational capability in separating PET, HDPE, and PP is tested to evaluate the functionality of the design. This ensures that stakeholder requirements are met. Although the testing and sampling is done in Malawi, the evaluation of the separation purity is done in Germany using [RTT's flakeanalyser](#), since no feasible in-line analysis method exists.

Description of Test

After the assembly of the basins and trays is finished (before applying the silicone), the trays are inserted into each of the two basins to scrutinize that the tolerances were met. This is done for each tray in all possible orientations, rotating the trays by 90° each time.

The leakage test is conducted after applying the silicone seal on the basins of the prototype. The basin with the cured silicone seals is placed on a dry surface and is filled with 50 liters of water. During the next five hours, the outside of the basin and the floor below is checked for any water drops.

Before the separation can be performed, the plastic flakes for the test must be prepared. For this, flakes from the stored plastic are used. Since the waste in the bags is not homogeneous, the content of multiple bags is taken to increase the mixture of different materials (use bags 1 and 4 from test 6 "[Plastic waste analysis - Blantyre, Malawi](#)"). A total of 200 kg is collected. Due to the poor integrity of the current bags in the storage, the flakes are transferred into new bags for transportation, each bag containing 50 kg of flakes. These flakes are then transported to Waste Advisers' Innovation Hub where the sink-float separation prototype is set up.

To prepare the flakes for the separation process, they are washed with detergent and rinsed with clean water. Add roughly 50 kg of flakes to a plastic drum with water and detergent at a time, stir the mixture for 2 minutes and drain the content of the drum in the mesh-lined container (green) to strain the washed plastic flakes from the water. Rinse the flakes before transferring them to the storage container for mixed plastic (green). Repeat the process to clean all flakes. Take a 1 l sample of the washed, unsorted plastic flakes.

For the plastic separation, fill basin 1 with 30 l of water and add a little bit of detergent to reduce the surface tension. Place tray 1 in the basin and add 10 kg of plastic flakes. Depending on the fill level, height adjustment of the stirrer holder might be necessary. The stirrer should be positioned just below

the water surface without touching the mesh of the tray. Then, use the stirrer to induce a current in the mixture to wet all plastic flakes well. After removing the stirrer from the basin and letting the flakes come to a rest, use the strainer to skim the floating flakes on the surface and transfer them into tray 2. When no more flakes float in the water, carefully lift the tray out of the water straining the sunken flakes. Transfer the sink fraction to the sorted PET container (orange) and take a 1 l sample. Fill basin 2 with 15 l of water and 15 l of ethanol. After mixing, take a sample of the solution with the measuring cylinder and test its ethanol content with the hydrometer. Adjust the mixture by adding ethanol or water to reach a solution at 50 vol %. Insert tray 2 with the float fraction from basin 1 into basin 2 and use the stirrer to wet all flakes. Remove the stirrer and, when the flakes are at rest, skim the floating flakes using the strainer and transfer the flakes to the sorted PP container (red). When the float fraction is removed, strain the remaining flakes by lifting tray 2 out of the solution and transfer the flakes to the sorted HDPE container (blue). Take a 1 l sample of the sorted PP and HDPE each.

After the testing is complete, the samples are transported to Germany. Here, the samples' composition was analyzed at the K 2025 ("World's No. 1 Trade Fair for Plastics & Rubber") in Duesseldorf, Germany, using the [flakeanalyser](#) from RTT.

Inside the device, the flakes are recorded from different angles by numerous (optical) sensors. To identify different plastic types, primarily a near-infrared (NIR) sensor is used, whose measured spectrum is matched with the characteristic spectra of the different materials. The superposition of these images results in outputs for object count, area, and volume (all absolute and relative) differentiated by color and material type. The mass fraction, i.e. purity, of the different materials is derived from the volume measurements. Additionally, statistics of the grain sizes in the sample are returned.

During this process the following constraints influence the results:

- Due to their reduced reflection, black or dark-colored objects cannot be classified.
- LDPE and HDPE are indistinguishable from each other due to similar spectra.
- The object count must be considered carefully, since the internal computations "cut" large pieces to decrease the miscounting of contiguous flakes. For this purpose, a maximum grain size is defined in the software (here, it was set to 10 mm), which is based on experience and the average flake size within the sample.
- Large and light pieces (e.g., long fibers) pose a risk to the precise measurements as they are prone to shifting during the measurement, influencing the superposition of the different images negatively.

Calculations

None.

Schedule of test with all steps required

Fit test:

1. Insert tray 1 fully into basin 1 and ensure there is sufficient space between the parts.
2. Repeat **step 1** three more times while rotating the tray by 90° in between.
3. Repeat **steps 1 and 2** with basin 2.

4. Repeat **steps 1-3** for tray 2 and tray 3.

Leakage Test:

1. Set up the basins on a dry, flat surface.
2. Fill the basins with **50 l** of water.
3. Check the floor and the outside of the basin for water droplets during the next 5 hours.

Separation test:

1. Prepare the new storage bags for transferring of the flakes in the storage.
2. Select a bag and transfer the content to a new bag.
3. Close the bag and load onto car.
4. Repeat **steps 2 and 3** for different bags/different plastic types until the target weight of **200 kg** is reached (bags 1 and 4 from test 6).
5. Transport the test flakes to the Innovation Hub.
6. Add **50 kg** of flakes to the plastic drum. Add water until the flakes are well wetted with enough room for the flakes to float freely. Add detergent to mix.
7. Stir with a stick for 2 min.
8. Carefully drain content of drum in batches through the green mesh-lined container.
9. Rinse flakes with water to clean off excess foam.
10. Transfer flakes to the storage container for mixed plastic flakes (green).
11. Repeat **steps 6-10** until all flakes are washed.
12. Take **1 l sample of the washed, unsorted flakes**.
13. Fill basin 1 with **30 l of water** and add **15 g of detergent**.
14. Place tray 1 in the basin.
15. Fill the tray with **10 l** of the mixed plastic.
16. Adjust the stirrer height, so that the stirrer head is positioned just below the water surface without touching the mesh of the tray.
17. Use the stirrer to stir the mixture to wet all flakes properly (**5 min**).
18. Remove the stirrer and let the flakes settle until all vertical movement has stopped.
19. Use the strainer to skim the float fraction. Transfer the flakes to tray 2.
20. When no floating plastic remains, lift tray 1 with the remaining sink fraction out of the basin, draining the water in the process. Transfer the flakes to the sorted PET container (orange).
21. (If not enough sink or float fraction for sample repeat **steps 14 – 20**.)
22. Take **1 l sample of the sorted PET** from the PET container.
23. Fill basin 2 with **15 l of water** and **15 l of ethanol** and mix.
24. Verify the ethanol content in the solution of **50 vol %** with the hydrometer. Adjust if needed.
25. Place tray 2 with the float fraction from basin 1 in basin 2.
26. Repeat **steps 16-18**.
27. Use the strainer to remove the float fraction from the basin and transfer them to the sorted PP container (red).
28. When no floating plastic remains, lift tray 2 with the remaining sink fraction out of the basin, draining the water in the process.
29. Transfer the sink fraction to the sorted HDPE container (blue).

30. Take 1 l sample of the sorted HDPE from the HDPE container.

31. Take 1 l sample of the sorted PP from the PP container.

Analysis of sample composition:

1. Follow the instructions by RTT.

Risk analysis

Potential risks associated with this test

The origin and content of the bags is unknown. Harmful dust particles could be present and dispersed in the air while sampling.

The bags have become very fragile over the long storing period and rip easily, possibly resulting in an avalanche like material stream.

Sharp objects could hide within the material. Small particles could get into the eyes.

Ethanol is highly flammable in liquid and vapor form and can cause serious eye irritation.

Mitigation steps

A face mask (FFP3 recommended, minimum FFP2), safety glasses, and gloves are worn during the sampling. Furthermore, high boots are worn to improve stability and protect the ankles from falling material. Additionally, acquiring the flakes is handled by two people.

The tests including ethanol are done in a ventilated area and possible sources of fire are removed.

Safety glasses are worn at any time during the testing with ethanol.

Risks after taking mitigation steps

Ripping of bags and loosening of large amounts of plastic flakes is still possible. The sampling must be handled with care.

Tools and devices required

Tool	Check-in tick
2x safety glasses	X
2x pair of gloves	X
2x face mask (min. FFP2, rec. FFP3)	X
2x pair of boots	X
1x key to storage	X
1x knife	X
4x new storage plastic bags	X

Duct tape	X
Dish soap	X
Water	X
1x large plastic drum for cleaning	X
1x stirrer stick	X
1x mesh-lined container	X
4x storage container for mixed plastic flakes and sorted plastics	X
Sink-float separation prototype	X
1 l measuring cup	X
1x hydrometer	X
2x strainer	X
20 l of ethanol	X
4x 1 l sample bags	X
1x laptop	X
1x phone (camera and watch)	X

Testing Results

All recordings and pictures of the process are available [here](#).

Fit Test – 08.09.2025

All trays could be inserted with ease in all orientations for both basins.

Leakage Test – 08.09.2025

No leakage could be detected during the five hour test time.

Addendum (22.9.25): After working on the prototype over the weekend at home outside, the silicone seal of one basin came off on two edges, possibly due to sunlight and/or heat. Due to time constraints, the seal cannot be redone before the prototype functionality testing (three days curing time). Instead, the seals were covered with duct tape to keep the silicone joints in place and reduce the friction between the edges of the trays and the silicone joints for the full prototype test. After the tests, the seal is renewed.

Functionality Test – 23.-24.09.2025

Plastic flakes acquisition from storage

Roughly 15 % of the bags in the storage were gone, including the bags sampled in test 6. It was unknown that the plastic is still used by MTEC, a company cooperating with Waste Advisers, for their projects.

The majority of remaining plastic (in accessible bags) was PET, while only half a bag of HDPE was found (maybe other material somewhere in the pile). Only two bags in total were collected and transported to the Innovation Hub for testing, since bringing more relatively pure PET did not add value to the test.

Other bags in storage not taken to testing also included large pieces of plastic foil.

Flake washing

Contamination found during washing: screw, metal pieces, ceramic tile pieces, charcoal, cement pieces (see Figure 1).



Figure 1: Contamination found with plastic flakes during washing.

Plastic separation by sink-float prototype

In contrast to the flake behavior observed in test 1 “[Sink-float separation - Proof of concept](#)” where work was required to wet the flakes, a large amount of flakes added to basin 1 in the first separation step immediately sank to the bottom, including flakes of a density lower than water that were pulled down by the surplus of PET flakes. This is likely a consequence of the preceding washing step, from which the flakes were still wet at the time of testing. Thus, two different separation procedures were tested:

1. One long period of stirring the flakes followed by skimming the floating fraction off the surface – suited for wetting hydrophobic plastic flakes.
2. Repeated execution of the stirring and skimming step. Here, a ladle was used instead of the stirrer for easier switching between the steps.

Both procedures were sampled and analyzed, resulting in a total of five samples. For the second sink-float process in basin 2, the second method was used as well, since its superiority for the present case was visually apparent.

Sample	Description	Ideal outcome
Sample 1	Basin 1, sink fraction – one long period of stirring	Pure PET
Sample 2	Basin 1, sink fraction – stirring and skimming multiple times	Pure PET
Sample 3	Basin 2, float fraction – stirring and skimming multiple times	Pure PP
Sample 4	Basin 2, sink fraction – stirring and skimming multiple times	Pure HDPE
Sample 5	Initial mixed plastic flakes	–

Sample Composition Analysis – 12.10.2025

The raw results’ data can be found [here](#).

For the analysis, the raw data underwent a series of data processing steps:

- Since the distinction between different colors of PE and PP is not significant, the individual measurements are summed up.
- ABS, PA, PC, and Other Plastics are grouped into Other Plastics due to the low number of detected flakes.
- The result is ordered to suit the topic of the thesis with the most important measurements on top.
- Lastly, the column mass [g] is removed since this value is not measured but calculated by the flakeanalyser based on the total weight of the sample. This value is not measured internally but entered by the user prior to the scan. An arbitrary value (123 g) was used in this case. Thus, the results are not significant.
- For clarity, only the most important values (mass [%], volume [%], and object count) are displayed in the derived results below.

The values for the material “Cell” cover cellulose-based materials such as paper or leaves. If flakes return no sufficient congruence to any entry in the database, it could either be due to their dark or black color – assigned to the material “Black” – or because no characteristic spectrum was recorded and saved for this material which is included as “Not identifiable”.

Sample 1

Basin 1, sink fraction – one long period of stirring

Ideal outcome: pure PET

Material	Mass [%]	Volume [%]	Object count []
PP	0.56	0.81	98
PE	6.90	9.74	680
PET	89.87	86.44	10,314
PS	0.00	0.00	1
PVC	0.45	0.43	49
Other Plastics	0.00	0.00	0
Metal	0.68	0.45	41
Cell	0.13	0.24	24
Not identifiable	0.00	0.00	0
Black	1.42	1.89	154

Sample 2

Basin 1, sink fraction – stirring and skimming multiple times

Ideal outcome: pure PET

Material	Mass [%]	Volume [%]	Object count []
PP	0.21	0.32	59
PE	0.35	0.51	83
PET	96.55	95.84	13,336
PS	0.00	0.00	0
PVC	0.76	0.75	83
Other Plastics	0.00	0.00	0
Metal	0.57	0.39	58
Cell	0.10	0.19	37
Not identifiable	0.00	0.00	0
Black	1.46	2.00	183

Sample 3

Basin 2, float fraction – stirring and skimming multiple times

Ideal outcome: pure PP

Material	Mass [%]	Volume [%]	Object count []
PP	69.76	70.46	5,641
PE	29.93	29.26	2,975
PET	0.03	0.02	37
PS	0.00	0.00	2
PVC	0.04	0.03	14
Other Plastics	0.00	0.00	5
Metal	0.00	0.00	0
Cell	0.02	0.03	32
Not identifiable	0.00	0.00	0
Black	0.20	0.19	18

Sample 4

Basin 2, sink fraction – stirring and skimming multiple times

Ideal outcome: pure HDPE

Material	Mass [%]	Volume [%]	Object count []
PP	0.10	0.11	16
PE	99.77	99.81	7,411
PET	0.10	0.07	8
PS	0.00	0.00	0
PVC	0.02	0.01	3
Other Plastics	0.00	0.00	0
Metal	0.00	0.00	0
Cell	0.00	0.00	2
Not identifiable	0.00	0.00	0
Black	0.01	0.01	2

Sample 5

Initial mixed plastic flakes

Material	Mass [%]	Volume [%]	Object count []
PP	2.58	3.52	406
PE	19.97	26.37	2,402
PET	75.21	67.64	11,390
PS	0.00	0.00	1
PVC	0.65	0.58	74
Other Plastics	0.00	0.00	2
Metal	0.23	0.14	31
Cell	0.11	0.20	55
Not identifiable	0.00	0.00	0
Black	1.25	1.55	137

Evaluation

The successful fit and leakage tests allowed to proceed with the manufacturing.

Although enough HDPE and PP flakes were available for the prototype testing, the remaining plastic bags in the storage contain PET with high purity. Despite not being designed for this case, the prototype can still be used to improve the purity further by removing remaining plastics like LDPE, HDPE, and PP in the water basin.

Pollutants such as tile or metal pieces found in the plastic are not addressed by the prototype. Large pieces can be picked out by hand, whereas an additional cleaning step using magnets or eddy currents could facilitate the removal of metal. Depending on the intended use of the separated flakes, this must be addressed in a follow-up project.

When evaluating the purity results of sample 1 and sample 2, the assumption from the visual assessment was confirmed: with a purity of 96.55 %, the process applied to sample 2 not only exceeds the purity of sample 1 but also meets the prototype's target of at least 95 %. Thus, repeated stirring and skimming is recommended for the present conditions in order to obtain PET at the desired purity.

Sample 3 exposed an insufficient outcome in PP separation. Possible explanations are a distortion of the result due to long PP fibers or the influence of LDPE in the initial plastic mix. The density of LDPE (0.924 g/cm^3) lies right within the targeted separation medium density (0.9265 to 0.9227 g/cm^3 , see test 4 “[Light sink float separation - Proof of concept](#)”). Only small variations in the chemical composition of the medium or ambient temperature can manipulate the allocation of LDPE to the float or sink fraction. Since the overall content of PP in the initial plastic mix is relatively small with only 2.58 % (see Sample 5), low quantities of LDPE assigned to the float fraction can strongly sway the outcome. Consequently, the PP can either be discarded or purified in an additional process

step.

Surprising are the presence of PVC and PET flakes in sample 3, since their densities are much higher than that of the separation medium. Possible explanations are multilayered plastics of different types or the interlocking of different flakes.

The highest purity of 99.77 % was recorded for PE (mostly HDPE) in sample 4, with only 29 to 31 out of 7442 flakes being assigned to the sink fraction incorrectly. This clearly exceeds the purity target and facilitates the reuse of the material. Depending on the designated recycling application, a possible presence of LDPE flakes in the material must be accounted for.

Despite reaching the purity goals for PET and HDPE, impurities detected in the samples can reduce reusability of the plastics. While not identified during test 6 “[Plastic waste analysis - Blantyre, Malawi](#)”), even small quantities of PVC can have negative impacts on the mechanical and optical properties of the recycled product (ASG, 2013). Depending on the intended use, this must be addressed in future work.

Remarks:

- Since black or dark flakes cannot be assigned by the flakeanalyser, the real results likely deviate slightly from the results presented above. However, these flakes are represented in a separate row, which leads to a conservative estimate of the result. The real purity value is equal to or better than the measured value.
- Classification by comparing spectra is an estimate, since the measured can deviate from the material’s characteristic spectrum (see the [RTT lab report](#)). However, the specific decision-making process behind the flakeanalyser is unknown. Thus, an ideal identification of materials is assumed as no statistical analysis is possible.

Bibliography

ASG. (2013, June 5). PVC in PET Bottle Recycling. *PET Bottle Washing Line | ASG Plastic Recycling Machine*. <https://www.petbottlewashingline.com/pvc-in-pet-bottle-recycling/>